



Local Agricultural Price & Climate Predictor Project Report

COMP-4202

Practical Data Science

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Approach and Methodology

Our study employs a hybrid analytical framework combining **Agricultural Economics** with **Time Series Forecasting**. The primary goal was to predict the `Fiyat_Degisim_Yuzdesi` (Percentage Price Change) of products in the Konya Wholesale Market.

- **Data Integration:** We merged two disparate datasets: historical market price sessions (2021-2026) and high-resolution climate data from the Open-Meteo API.
- **Inflation Normalization:** To ensure the model learned real market volatility rather than nominal currency fluctuations, prices were adjusted using local inflation indices.
- **Time Series Validation:** Unlike standard cross-validation, we used `TimeSeriesSplit` and a final **30-day Hold-out Test Set** to maintain chronological integrity and prevent data leakage.

Model Selection and Justification

We benchmarked statistical algorithms against ensemble Machine Learning (ML) models.

- **Selected Model:** SARIMA(1,1,1)(1,1,1,2).
- **Justification:** While Prophet provided excellent interpretability and ML models (Random Forest) captured complex non-linear relationships, SARIMA excelled at modeling the **bi-weekly market rhythm**. Since the market operates specifically on Mondays and Thursdays, the seasonal order of $s=2$ allowed the model to focus on session-to-session volatility, outperforming others in RMSE during back-testing.

Results and Evaluation Metrics

Model performance was evaluated across three back-testing windows to ensure stability.

- **RMSE (Root Mean Squared Error):** This was our primary metric for selection as it heavily penalizes large errors, which is vital for detecting sudden price shocks.
- **MAPE (Mean Absolute Percentage Error):** Our target was $\% \leq 5$. The final MAPE achieved by SARIMA on the test set suggests a robust generalization capability.
- **wQuantileLoss (P50):** Applied as per Amazon Forecast standards, this metric validated that our median forecast successfully represents the central tendency of the market.

Key Findings and Literature Comparison

Our findings align with recent studies on "Climate-Smart Agriculture," confirming that localized weather events (frost and extreme heatwaves) are leading indicators of short-term price volatility in Mediterranean climates like Konya. Unlike many global studies that use monthly data, our **session-based analysis** provides higher precision for local farmers and stakeholders. The probabilistic forecast (P10/P90) offers a risk-management tool that deterministic models lack, mirroring the output quality of professional cloud solutions like Amazon Forecast.

Challenges and Solutions

The Panel Data Dilemma: Time series models like SARIMA typically handle single series. Our dataset had multiple products (Panel Data).

- *Solution:* We aggregated product-level data into a market-average series for statistical modeling while retaining granular features for ML models.

Seasonality Complexity: Traditional daily seasonality ($s=7$) failed due to market opening days.

- *Solution:* We re-engineered the seasonality index to focus on the session count per week ($s=2$), significantly improving the autocorrelation (ACF) results of the residuals.

Discussion and Literature Comparison

1. Interpretation of Model Performance

The empirical results of this study demonstrate that **SARIMA(1,1,1)(1,1,1,2)** outperformed more complex Machine Learning (ML) architectures such as Random Forest and Gradient Boosting. This finding aligns with the "Principle of Parsimony" (Occam's Razor) in time series forecasting: when the data exhibits strong, rigid seasonality (the bi-weekly market cycle), statistical models often provide more robust generalization than ML models, which may struggle with the "small data" problem in niche local markets.

2. Comparison with Existing Literature

Current literature on agricultural price forecasting (e.g., studies by Zhang et al. on vegetable price volatility) often suggests that hybrid models or Deep Learning (LSTM) are superior. However, those studies typically focus on large-scale national indices with daily or monthly observations. Our research offers a different perspective: for **localized wholesale markets** with specific session constraints (Mondays and Thursdays), the seasonal integration in SARIMA ($s=2$) is more effective at capturing short-term mean reversion than non-linear ML models.

3. The Climate-Price Nexus

Our findings confirm the significant impact of extreme weather events—specifically frost—on price spikes, a correlation well-documented in agricultural economics literature. By integrating Open-Meteo climate variables as exogenous

factors (even as lag features for ML), we validated that localized high-resolution weather data increases the explainability of price shocks compared to models using only historical price data.